



Curriculum Vitae

Gustavo Pinho Maia

I am a Ph.D. student and researcher at the Centro de Química Estrutural (CQE), working in collaboration with the Muséum National d'Histoire Naturelle (MNHN) in Paris and the Astrobiology Analytical Laboratory at NASA's Goddard Space Flight Centre. My research lies at the interface of prebiotic chemistry, mechanochemistry and meteoritics, aiming to understand how organic molecules originated, transformed and persisted across planetary environments. I investigate how mechanical energy, generated during parent-body formation, impact processes or meteoritic atmospheric entry, can drive the synthesis, degradation and evolution of organic compounds relevant to the emergence of life.

My scientific path began in 2013 in mechatronics and robotics, where I cultivated a strong experimental and problem-solving mindset before transitioning to chemistry. I completed a BSc in Industrial Chemistry at the University of Beira Interior (2020), specialising in organic synthesis and optimising routes to quinoline N-oxides for pharmaceutical applications. I later earned my MSc in Chemistry at Instituto Superior Técnico (2022), focusing on meteorite analysis, prebiotic chemistry and mechanochemical reactivity. Throughout my academic training, I have presented my research at national and international conferences — including the XV Annual CICS-UBI Symposium, the CQE Days. and EuChemS Chemistry Congress.

Alongside research, I actively engaged in scientific outreach and academic community building. I have participated in and co-organized events that promote the discussion of chemistry in Portugal, most notably the Digital Colloquium of Chemistry Students in Portugal (2021), which brought together distinguished national and international speakers.

Identification

Personal identification

Full name

Gustavo Pinho Maia

Gender

Male

Birth date

1998/09/22

Citation names

Maia, Gustavo P.

Maia, Gustavo

Author identifiers

Ciência ID

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ORCID iD

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Avenida Fernão Magalhães, 363, 3880-354 Furadouro, Ovar, Portugal (Personal)

Websites

https://www.researchgate.net/profile/Gustavo_Maia2 (Professional)
www.linkedin.com/in/gustavopinhomaia (Social media)

Knowledge fields

Exact Sciences - Chemical Sciences - Organic Chemistry

Languages

Language	Speaking	Reading	Writing	Listening	Peer-review
Portuguese (Mother tongue)					
English	Upper intermediate (B2)	Upper intermediate (B2)	Intermediate (B1)	Advanced (C1)	Upper intermediate (B2)

Education

	Degree	Classification
2023/12/01 - 2027/12/01 Ongoing	Chemistry (Doutoramento) Major in Astrobiology Universidade de Lisboa Instituto Superior Técnico, Portugal <i>"Mechanochemical energy and prebiotic synthesis"</i> (THESIS/DISSERTATION)	
2020/09 - 2022/12 Concluded	Química (Mestrado) Major in Química Universidade de Lisboa Instituto Superior Técnico, Portugal <i>"Ribonucleoside stability and prebiotic synthesis"</i> (THESIS/DISSERTATION)	17
2017/09 - 2020/09 Concluded	Química Industrial (Licenciatura) Universidade da Beira Interior, Portugal <i>"Otimização da síntese de N-óxidos de quinolina a partir da redução da 2-(2-nitrobenzilideno)ciclohexano-1,3-</i>	16

*diona"" (THESIS/DISSERTATION)*2013/09 - 2016/09
ConcludedCurso Profissional Técnico de Mecatrónica Nível IV (Ensino secundário)
Major in Robótica

18

Escola Profissional de Espinho, Portugal

Affiliation**Teaching in Higher Education**

Category

Host institution/organization

Employer

2024/09/01 -
2024/10/31Invited Assistant (University Teacher)
Universidade de Lisboa Instituto
Superior Técnico, PortugalUniversidade de Lisboa Instituto
Superior Técnico, Portugal**Positions / Appointments**

Category

Host institution/organization

Employer

2019/10/10 -
2020/09/20Delegate of Industrial Chemistry
Students at the Pedagogical Council of
the University of Beira Interior
Universidade da Beira Interior, Portugal

Universidade da Beira Interior, Portugal

Others

Category

Host institution/organization

Employer

2025/01/10 - Current

Visiting Scientist
Institut de minéralogie, de physique des
matériaux et de cosmochimie, FranceMuséum National d'Histoire Naturelle,
France

2024/11/09 - 2024/12/01

Visiting Scientist
NASA Goddard Space Flight Center,
United StatesNASA Goddard Space Flight Center,
United States

2021/06/12 - 2022/05/12

President of the Fiscal Council for the
National Association of Chemistry
Students
Universidade de Coimbra Faculdade de
Ciencias e Tecnologia, PortugalAssociação Nacional de Estudantes de
Química, Portugal

2020/06/05 - 2021/06/05	Vice President of the National Association of Chemistry Students (ANEQ) Universidade de Coimbra Faculdade de Ciencias e Tecnologia, Portugal	Associação Nacional de Estudantes de Química, Portugal
2019/09/12 - 2020/12/15	Vice-President of the Student Group of Chemistry and Biochemistry at the University of Beira Interior (UBIQUIMICA) Universidade da Beira Interior, Portugal	UBIquímica, Portugal
2020/07/22 - 2020/09/11	Research Intern Universidade da Beira Interior, Portugal	Universidade da Beira Interior, Portugal
2019/08/01 - 2019/09/04	Factory worker Nestle Portugal S A, Portugal	Randstad, Portugal
2018/08/28 - 2019/08/28	Member of the Scientific Committee of the Group of Students of Chemistry and Biochemistry at the University of Beira Interior (UBIQUIMICA) Universidade da Beira Interior, Portugal	UBIquímica, Portugal
2016/08/02 - 2016/09/19	Cafe employee	Parque de Campismo do Furadouro, Portugal
2016/04/20 - 2016/07/29	Professional Internship Escola Profissional de Espinho, Portugal	Escola Profissional de Espinho, Portugal
2015/06/10 - 2015/07/27	Professional Internship Escola Profissional de Espinho, Portugal	Escola Profissional de Espinho, Portugal
2014/06/10 - 2014/07/27	Professional Internship Ramada Aços, S.A, Portugal	Ramada Aços, S.A, Portugal

Projects

Grant

	Designation	Funders
2023/12/01 - 2027/11/30	Mechanochemical energy and prebiotic synthesis 2023.01099.BD 10.54499/2023.01099.BD Principal investigator Universidade de Lisboa Instituto Superior Técnico, Portugal	Fundação para a Ciência e a Tecnologia, Portugal Ongoing

Contract

	Designation	Funders
2020/01/01 - 2024/12/31	Centro de Química Estrutural UIDB/00100/2020 10.54499/UIDB/00100/2020 Instituto Politécnico de Setúbal, Portugal Instituto Politécnico de Lisboa Instituto Superior de Engenharia de Lisboa, Portugal Universidade de Lisboa Instituto Superior Técnico, Portugal Universidade de Lisboa Centro de Química Estrutural, Portugal Universidade de Lisboa Faculdade de Ciências, Portugal Associação do Instituto Superior Técnico para a Investigação e Desenvolvimento, Portugal FCiênciasID Associação para a Investigação e Desenvolvimento de Ciências, Portugal	Fundação para a Ciência e a Tecnologia, Portugal Concluded
2023/06/01 - 2023/11/30	A Gold Nanosensor for Single Molecule Detection of RNA Biomarkers in a Miniaturized Microscope 2022.04076.PTDC Research Fellow Universidade de Lisboa Centro de Química Estrutural, Portugal	Fundação para a Ciência e a Tecnologia, Portugal

Outputs

Publications

Conference poster	1	Maia, Gustavo P.; Silva, José; K Amano; J-C Viennet; Dworkin, Jason P.; McLain, Hannah; Laurent, Remusat. Corresponding author: Maia, Gustavo P.. "Solid-state mechanochemistry of extraterrestrial organic molecules: new insights on the exogenous delivery of organic matter". Paper presented in <i>EANA 2025</i> , 2025.
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- 2 Maia, Gustavo; José Armando Luísa Da Silva; Dworkin, Jason P.; Remusat, Laurent. Corresponding author: Maia, Gustavo. "Shock-Impact Events and the Origin of life: a mechanochemical approach". Paper presented in *PhD Open Days 2024*, 2024.
- 3 Maia, Gustavo; Adelino Galvão; José Armando Luísa da Silva. Corresponding author: Maia, Gustavo. "Ribonucleosides stability and prebiotic synthesis". Paper presented in *CQE DAYS 2022*, 2022.
- 4 Maia, Gustavo; Paulo Jorge Silva Almeida; Silvestre, Samuel; João Paulo Lourenço Serrano. Corresponding author: Maia, Gustavo. "PREPARATION OF QUINOLINE N-OXIDES BY 2-NITROBENZYLIDENE REDUCTION: AN OPTIMIZATION STUDY". Paper presented in *XV ANNUAL CICS-UBI SYMPOSIUM 2020*, 2020.

Journal article	1	Gustavo P. Maia; Catarina Gonçalves; Ana J. Carvalho; Vânia André; Adelino Galvão; Ana P. C. Ribeiro; Pedro F. Pinheiro; José Armando Luísa da Silva. "Mechanochemical Reactivity of Ribonucleosides Mediated by Inorganic Species: Implications for Extraterrestrial Organic Matter Interpretation". <i>Applied Sciences</i> (2025): https://doi.org/10.3390/app15031363 . 10.3390/app15031363
	2	Gustavo P. Maia; José Armando Luísa da Silva; Vânia André; Adelino M. Galvão. "Shock-Induced Degradation of Guanosine and Uridine Promoted by Nickel and Carbonate: Potential Applications". <i>Molecules</i> (2023): https://doi.org/10.3390/molecules28248006 . 10.3390/molecules28248006
	3	Ana Franco; Gustavo P. Maia; J. A. L. da Silva. "Da Química Bioinorgânica para a Química Prebiótica: Levar Mais Longe o Legado do Prof. Fraústio da Silva". <i>Boletim da Sociedade Portuguesa de Química</i> 47 171 (2023): 49-240. http://dx.doi.org/10.52590/m3.p708.a30002745 . Published · 10.52590/m3.p708.a30002745
Thesis / Dissertation	1	"Ribonucleoside stability and prebiotic synthesis". Master, Universidade de Lisboa Instituto Superior Técnico, 2022. https://fenix.tecnico.ulisboa.pt/cursos/mq/dissertacao/565303595503245 .

Activities

Oral presentation

	Presentation title	Event name Host (Event location)
2025/10/20	The effect of mechanochemistry on the formation of extraterrestrial organic matter	AbGradE'25 Symposium Astrobiology Graduates in Europe (Lisbon, Portugal)

2025/10/15	Mechanochemistry and the fate of extraterrestrial organic matter	"Small Bodies" Day Symposium Institut de physique du globe de Paris (Paris, France)
2025/07/02	Effect of Mechanochemical Events Into Extraterrestrial Organic Matter and Its Implications for Exogenous Delivery	BEACON 2025 European Astrobiology Institute (Reykjavik, Iceland)
2025/06/23	The effect of shock/impacts on the exogenous delivery of organic matter	Institut de minéralogie, de physique des matériaux et de cosmochimie PhD Students' s Day IMPMC (Paris, France)
2024/10/28	Impact events of extraterrestrial bodies: The bridge between exogenous delivery and the origin of life	NInTec Science Days 2024 NInTec - Núcleo de Investigadores (Lisbon, Portugal)
2022/08/29	Why not nucleosides in meteorites? An approach based on mechanochemical studies	8thEuChemS Chemistry Congress (ECC8) Portuguese Chemical Society (SPQ) (Lisbon, Portugal)

Event organisation

	Event name Type of event (Role)	Institution / Organization
2021/06/01 - 2022/03/31	+FUTURO - This activity aims to communicate to undergraduate students the different master's degrees in chemistry and related areas, through clarification sessions with coordinators, students and alumni of the respective master's degree. (2022/03/01 - 2022/03/31) Meeting (Co-organisator)	Associação Nacional de Estudantes de Química, Portugal
2020/06/15 - 2021/03/28	Co-organizer of the Digital Colloquium of Chemistry Students in Portugal 2021, which brought together speakers of recognized national and international merit. The scientific program included several thematic communications, with the objective of bringing chemistry closer to the scientific community and the general public. (2021/03/26 - 2021/03/28) Meeting (Co-organisator)	Associação Nacional de Estudantes de Química, Portugal
2019/04/06 - 2019/04/06	Organizing member of the Junior Chemistry Olympiad 2019 semifinals, held at the Chemistry Department of the University of Beira Interior (2019/04/06 - 2019/04/06)	Universidade da Beira Interior, Portugal

Call (Co-organisator)

2018/11/01 - 2019/04/04	Co-Organization of the 6th Conference on Chemistry and Biochemistry, held by the UBI Chemistry and Biochemistry Students Group (UBIquímica), in Covilhã, Portugal (2019/04/03 - 2019/04/04) Meeting (Member of the Organising Committee)	Universidade da Beira Interior, Portugal
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Event participation

	Activity description Type of event	Event name Institution / Organization
2025/03/17 - 2025/03/21	"RED is a Training School about the "Basics in Astrobiology". Astrobiology addresses questions related to the origins of life, its evolution, and distribution in the Universe. This school aims to offer an interdisciplinary training in astrobiology to students and young researchers having a master's degree in astronomy, planetology, geology, chemistry, biology, or history and philosophy of science." - Rencontres exobiologiques pour doctorants - An Astrobiology Introductory Course Conference	Rencontres exobiologiques pour doctorants - An Astrobiology Introductory Course
2016/06/30 - 2016/07/04	RoboCup Competition (Junior Rescue League) in Leipzig, Germany	RoboCup 2016 RoboCup Federation, United States
2016/05/04 - 2016/05/08	XVI National Robotics Festival (Junior League - Search and Rescue modality), in Bragança, Portugal Other	XVI National Robotics Festival Instituto Politécnico de Bragança, Portugal
2015/04/08 - 2015/04/12	XV National Robotics Festival (Junior League - Search and Rescue modality), in Vila Real, Portugal Other	XV National Robotics Festival Universidade de Trás-os-Montes e Alto Douro, Portugal
2014/05/14 - 2014/05/18	XIV National Robotics Festival (Junior League - OnStage and Search and Rescue categories), in Espinho, Portugal Other	XIV National Robotics Festival Escola Profissional de Espinho, Portugal

Interview (newspaper / magazine)

	Activity description	Newspaper / Forum
2025/09/10	Migrações - Ciência, Cultura e produção de conhecimento além fronteiras: O percurso improvável de Gustavo Maia levou-o de Ovar até à NASA	Maré Viva - Jornal Regional de Espinho
2024/12/04	Gustavo Maia aprofunda investigação no Laboratório Analítico de Astrobiologia da NASA	Praça Pública
2024/11/13	"Portanto, a Vida foi criada a partir de surras." – Entrevista com Gustavo Maia	Diferencial - Student Journal of IST

Distinctions

Award

2026	Excellent Teachers 2024/2025 Universidade de Lisboa Instituto Superior Técnico, Portugal
2025	Poster Award EANA 2025
2016	Rotary Club Merit Recognition Diploma Rotary Club de Espinho, Portugal

Other distinction

2016	CERTIFICATE OF AWARD - Best Presentation RoboCup Federation, United States
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